

Scalability & Future Plan

Date	10 JULY 2024
Team ID	XXXXXX
Project Name	Rising Waters – Flood Prediction ML Model
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Model uses historical data only	Predictions may not reflect real-time weather conditions	High
2	No real-time weather API integration	Users cannot receive live flood predictions	High
3	Prototype is designed for limited users	Performance may decrease with large-scale usage	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports only a few simultaneous users	Deploy on cloud platforms (AWS, Azure, GCP) with auto-scaling
2	Data Storage	Local dataset stored on system	Migrate to cloud databases for efficient storage and backup
3	Performance	Model runs on local machine	Optimize model, use GPU/cloud inference, caching, and load balancing
4	Security	Basic local access	Implement user authentication, encrypted APIs, and secure cloud deployment

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Improve UI and user experience	1 Month	Better usability
Phase 2	Integrate live weather API	2 Months	Real-time flood prediction
Phase 3	Add GIS maps and mobile application	3-4 Months	Better visualization and accessibility
Phase 4	SMS/Email alerts and Government integration	6 Months	Faster emergency response and wider adoption